# Firm Characteristics and the Stock Price Response to Fed Announcements

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**Abstract**

When the Federal Reserve announces a decision, some stock prices move sharply and others barely move. We ask why, using prices for S&P 500 firms in a 30-minute window around FOMC announcements from 2004 to 2024. Following the literature, we distinguish three kinds of surprise inside a single announcement: a shift in policy that economic conditions do not explain, news the Fed reveals about the economy, and news about how strongly the Fed will react to incoming data. Which firms move most depends on which surprise is at work. Growth firms, whose value comes from profits expected far in the future, gain most when the Fed signals the economy is strong, and lose most when the Fed signals it will tighten more aggressively in response. Firms whose profits track the economy gain more than others when the Fed reveals good news about the economy. How much debt a firm carries makes little reliable difference. The differences are small but systematic: within half an hour, markets sort firms by when their profits arrive and how much those profits depend on the economy, not by how much they owe.

# 1 Introduction

When the Federal Reserve surprises markets with a change in monetary policy, the ripple is sent through equity markets, but not uniformly across firms. Why do some firms' stock prices reflect greater changes during surprises, while other firms do not budge? This paper examines how firms' characteristics shape the stock price reaction to the surprise component of announcements from the Federal Reserve.

To answer this question, we use high-frequency intraday stock price returns from S&P 500 firms from TAQ via the WRDS database, measured over a 30-minute window bracketing Federal Open Market Committee (FOMC) announcements. The tight window isolates the response of stock prices attributable to policy announcement only, under the standard high-frequency identification assumption that no other systematic news moves prices within the window. We merge this dataset with identified monetary policy shocks from Jarociński and Karadi [2025] and Bauer and Swanson [2023], as well as with firm-level characteristics from Compustat.

This data allows us to analyze the effects of a given monetary policy shock on the stock prices. We specifically look at how the transmission of policy shocks is affected by standard firm-level characteristics. The main empirical specification we use is panel regressions with high-frequency stock returns as dependent variable and announcement-level monetary policy shocks, together with interaction terms of these shocks with firm-level characteristics as independent variables. This allows us to measure how leverage, size, Tobin's q and other firm characteristics amplify or dampen the stock price reaction to the unexpected changes of monetary policy.

Bernanke and Kuttner [2005] established how Fed rate cuts lead to significant increases in stock prices. Similarly, Ozdagli [2018] and Gurkaynak et al. [2022] showed how firm characteristics and debt structure shape the reaction in stock prices. In addition, we can draw from Nakamura and Steinsson [2018] who demonstrate that in the 30-minute window around the FOMC announcements, markets react to the Fed's signals about the economy not just the rate change. This paper contributes by analyzing the intraday stock price reaction around FOMC announcements and how it varies across identified policy shocks.

Prior work establishes that firm characteristics shape the equity response to monetary surprises, but usually treats the surprise as a single object. We instead decompose the surprise into separate components, following established procedures in the literature [Jarociński and Karadi, 2025, Bauer and Swanson, 2023], and track the effect of each of those components on stock prices of individual companies. For example, one separates the change in policy from the news the Fed reveals about the economy — two parts that can push stock prices in opposite directions: when the Fed unexpectedly raises rates, stocks fall if the move is just tighter policy, but rise if it also signals a stronger economy. We then ask how stock prices respond to each component, how that response depends on firm-level characteristics.

We found that firm characteristics shape firms' stock price reactions to monetary policy shocks, however the results are modest. A firm one standard deviation higher in a given characteristic reacts roughly 2–8% more strongly than it does at its own historical average. The two most robust findings in our research are the interaction term of cyclicality under the CBI shock and Tobin's q under the FRN shock, both of which hold across all three specifications of within-firm, levels and within-sector. In contrast, the interaction effect of leverage is weak and does not hold consistently across all three specifications.

# 2 Data

## 2.1 S&P 500 companies

We measure stock surprises as the percentage change in a firm's stock price 10 minutes before and 20 minutes after the FOMC announcement to complete the 30-minute window. The stock price at a given moment of time is measured as a midpoint between best bid and best offer quotes[1]. We examined the S&P 500 index from early 2004 to 2024, with firms leaving and entering the index throughout this period. Figure 1 shows the number of unique firms in the sample per FOMC announcement over time. The number of firms in the sample is relatively stable, ranging from 435 to 455 per announcement throughout the sample period.

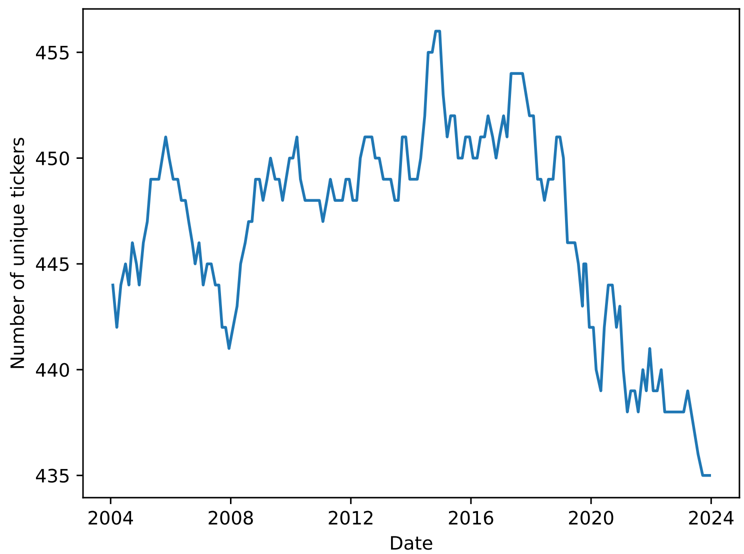


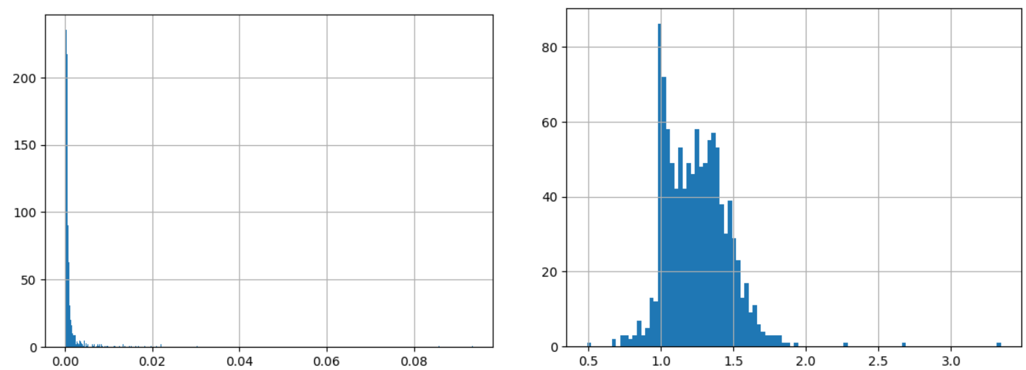

Figure 1: Number of Firms Per FOMC Announcement

The sample consists of 1,108 firms. Figure 2(a) analyses the spread before the FOMC announcement, with a median bid-ask spread of 0.0005.[2] Figure 2(b) shows the post/pre announcement spread ratio, where only 11.3% of firms have a post/pre spread ratio above 1.5. Even a ratio of 1.5 still implies a near-zero post-announcement spread, meaning liquidity concerns should

[1]There is a tiny number of instances where best bid and/or best offer is either unknown or zero - we drop such observations from the sample.

[2]The bid-ask spread is calculated as spread $= \frac{BO-BB}{(BO+BB)/2}$, where $BO$ is the best offer and $BB$ is the best bid.

not drive the forthcoming results.




(a) Bid-ask spread before FOMC announcement

(b) Post to pre-announcement spread ratio

Figure 2: Liquidity of firms

## 2.2 Policy shocks

A monetary policy surprise is the part of a policy decision that financial markets did not anticipate. The anticipated portion of a rate change is already incorporated into prices before the meeting, so the realized change in policy rate is a poor proxy for the new information the announcement conveys. Only the unanticipated component — the surprise — moves prices at the announcement. High-frequency identification literature measures this surprise from the change in interest-rate futures (or Eurodollar futures) over a short window around the announcement. Because the window is narrow, the revision in expected policy over that window can be attributed to the announcement itself rather than to other news arriving that day.

The surprise, however, is a reduced-form object: it measures the unanticipated change in monetary policy but does not identify its source. The same surprise can combine several distinct shocks — for example, an exogenous shift in the policy stance, due to unusual factors, new and unexpected information the Fed conveys about the economic outlook, or shift in the systematic reaction of the Fed to recent data that markets had underestimated.

Our main policy shocks come from Jarociński and Karadi [2025], who split the high-frequency Fed surprise into three parts:

- **Monetary policy (MP) shock** — a pure change in policy, akin to the residual of the policy rule. In response to this shock, interest rates and stock prices move in opposite directions: a surprise tightening raises rates and lowers stocks.

- **Central bank information (CBI) shock** — the Fed reveals its own view of the economy, that is different from the market's view. In response to this shock, interest rates and stocks move together: a tightening with rising stocks signals the Fed is more optimistic about the future outlook than the market.

- **Fed response to news (FRN) shock** — the market misjudged how strongly the Fed reacts to recent news. This part is predictable from public data.

We split the surprise this way because the three parts can move stock prices differently, and firms may respond to each of these shocks in different ways. For example, a high-growth firm might gain from a CBI shock that signals a stronger economy, but lose from an FRN shock that implies a more aggressive future rate path. A highly leveraged firm might react more strongly to an MP shock, because a surprise rate increase directly raises its cost of servicing and rolling over debt — a channel that barely touches a firm with little debt.

As a robustness check, we also look at propagation of two other shocks: the orthogonalized surprise of Bauer and Swanson [2023], which removes the predictable "response to news" part — the same channel Jarociński and Karadi [2025] isolate as the FRN shock — from monetary policy surprise, and the Fed non-yield shock of Boehm and Kroner [2024], which captures announcement-driven moves in stock prices that are unrelated to interest rates.

## 2.3 Firm-level characteristics

We select three firm characteristics (Leverage, Tobin's Q, Cyclicality), each motivated as the natural proxy for a distinct transmission channel, plus two standard controls. **Leverage**, computed as total (long- plus short-term) debt over total assets, captures the financial-frictions channel: more-indebted firms face steeper marginal financing costs and so respond differently to pure policy shocks. **Tobin's Q**, computed as (book assets + market equity - book equity) / book assets. If Tobin's $Q > 1$, the market values the firm at more than its assets are worth on paper, meaning investors expect those assets to generate returns above their cost - that's a growth firm with valuable future investment opportunities. So a high Q flags a firm whose value rests on future cash flows rather than assets already in place. **Cyclicality** measures a firm's cash-flow exposure to the macroeconomy: it is the slope from a rolling 20-quarter regression (minimum eight observations) of the firm's year-over-year ROA growth on real GDP growth, so that high-beta firms benefit disproportionately when the Fed signals favorable economic news. We additionally control for size, the log of total assets, and the pre-announcement bid-ask spread, the quoted spread relative to the midpoint, which absorb differences in firm scale and trading liquidity that could otherwise contaminate the high-frequency responses[3]. Table 1 shows basic descriptive statistics of the firm characteristics.

[3] For example, an illiquid firm's stock price may move less within the 30-minute window simply because trades arrive infrequently and prices adjust slowly, not because the firm is genuinely less sensitive to the policy news revealed at the announcement. Absent a liquidity control, this mechanical under-reaction would be misread as low policy sensitivity; including the bid–ask spread separates true sensitivity from sluggish price discovery.

Table 1: Descriptive statistics for firm-level characteristics.

| | Mean | SD | Min | p25 | Median | p75 | Max |
|---|---|---|---|---|---|---|---|
| Leverage | 0.270 | 0.166 | 0.000 | 0.156 | 0.257 | 0.373 | 0.955 |
| Tobin's $Q$ | 2.460 | 1.686 | 0.548 | 1.431 | 1.950 | 2.864 | 35.614 |
| Cyclicality | 4.117 | 21.890 | -183.083 | -3.456 | 1.786 | 9.373 | 200.328 |
| Bid–ask spread (%) | 0.098 | 1.277 | -0.127 | 0.024 | 0.038 | 0.063 | 199.674 |
| Size (log assets) | 9.430 | 1.147 | 5.785 | 8.578 | 9.353 | 10.177 | 13.649 |

Note: 53,769 firm–announcement observations; financials and utilities excluded.

# 3 Results

We start by considering the October 28–29, 2008 meeting of the Federal Open Market Committee, which took place at the height of the global financial crisis. Lehman Brothers had failed six weeks earlier, short-term funding markets had seized, and the Committee had already delivered an emergency, internationally coordinated 50 basis point cut on October 8. At the October 29 meeting, the FOMC cut the federal funds target by a further 50 basis points to 1%—its lowest level since 2004 and within reach of the zero lower bound it would hit that December—and the accompanying statement emphasized intensifying financial strains and mounting downside risks to growth[4]. We open with this announcement because it carries one of the largest identified monetary policy surprises in our sample and falls in a period of acute uncertainty, when differences across firms in their sensitivity to policy are most sharply revealed.

Figure 3 plots the cross-section of 30-minute stock returns across the 449 S&P 500 firms in our sample on that day, and its message is the contrast between a single common cause and a wide range of effects. Every firm in the figure was hit by the *same* announcement, at the *same* instant, within the *same* thirty-minute window: there is no cross-sectional variation in the

[4]`https://www.federalreserve.gov/newsevents/pressreleases/monetary20081029a.htm`

shock itself. Yet the price reactions are strikingly heterogeneous, spanning from below -4% to above +3%, with a cross-sectional standard deviation of 1.02%. Because the shock is identical across firms, this dispersion cannot reflect firms receiving different news—it must reflect differences in how firms are *exposed* to the same news. Identifying the firm characteristics that govern this differential exposure is the central task of the paper.

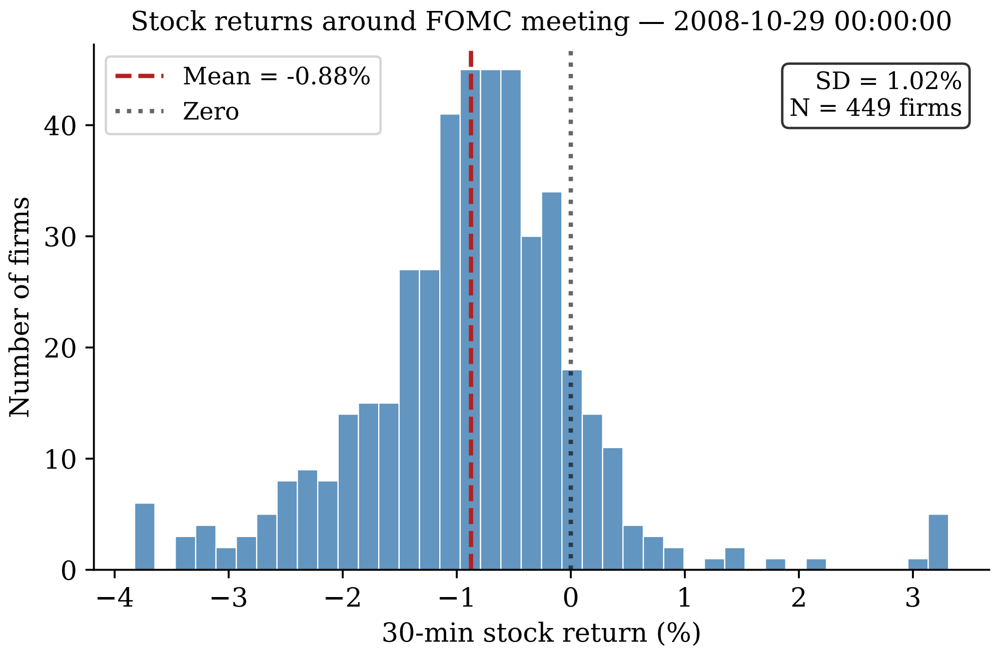



Figure 3: Cross-sectional distribution of 30-minute S&P 500 stock returns around the FOMC announcement of October 29, 2008 (a 50 bp cut to 1%). Despite the easing, the average firm declined by 0.88%, and responses are widely dispersed (SD = 1.02%, $N = 449$).

The evidence in Figure 3 that firms react differently to the same announcement is observed for the majority of other FOMC announcements in 2004–2024. For each FOMC announcement from 2004 to 2024, Figure 4 shows the cross-sectional distribution of individual stock price changes in the 30-minute window: the median across firms (black), the 25–75% range (purple), and the 5–95% range (grey). The median stays near zero, so the average stock barely moves; the informative feature is the dispersion across firms, which widens sharply in stressed periods (2008, 2020, 2022). This time-varying heterogeneity in responses to a common shock motivates the paper's focus on how firm characteristics shape stock price reactions.

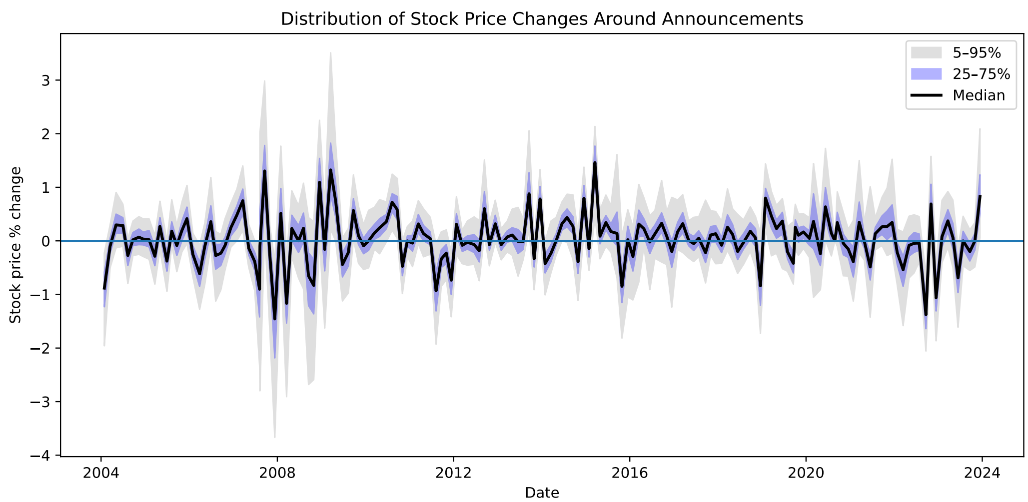



Figure 4: Distribution of stock price changes

## 3.1 Empirical specification

The dispersion in Figure 4 is systematic, not random: it shows up around nearly every announcement, so it reflects the policy news itself. This sharpens the question from Figure 3: if every firm faces the same surprise, why do some respond more than others? The natural answer is that firms differ in their exposure to monetary policy—through their leverage, the cyclicality of their cash flows, and how much of their value lies in future growth. The rest of this section shows that they do, and that which characteristic matters depends on which component of the surprise is moving.

We estimate how the high-frequency response of stock prices to monetary policy varies across firms and across the components of the policy surprise.

Our baseline specification is the firm-level panel regression

$$r_{i,t} = \alpha_i + \sum_k \beta_k \, S_{k,t} + \sum_k \sum_c \gamma_{c,k} \left( S_{k,t} \times \tilde{x}^c_{i,t} \right) + \varepsilon_{i,t}, \qquad (1)$$

where $r_{i,t}$ is the percentage change in firm $i$'s stock price in the 30-minute window around FOMC announcement $t$; $S_{k,t}$ is the surprise associated with component $k \in \{\mathrm{MP}, \mathrm{CBI}, \mathrm{FRN}\}$, normalized to unit standard deviation; and $\tilde{x}^c_{i,t}$ is firm characteristic $c$, winsorized (at the 0.5% and 99.5% levels), demeaned within firm, and standardized. The characteristics of interest are leverage, Tobin's $Q$, and cyclicality, with size and the bid–ask spread included as controls. The terms $\alpha_i$ are firm fixed effects, and standard errors are clustered by announcement date. The coefficients of interest are the interactions $\gamma_{c,k}$, which measure whether the price response to component $k$ depends on characteristic $c$.

Following Ottonello and Winberry [2020], we interact each shock with the characteristic measured *relative to the firm's own average*, $\tilde{x}^c_{i,t}$, rather than its raw level. This distinguishes two things that a level would mix together: whether a firm reacts more when it is temporarily more leveraged than usual (a *state* effect), versus whether some firms are simply always more sensitive to policy (a *type* effect). By removing each firm's own average, $\gamma_{c,k}$ captures the first—how a firm's response changes as its leverage, value, or cyclicality moves over time—rather than fixed differences across firms that may reflect many other traits.

The defining feature of (1) is that the policy surprise enters as three separate components rather than a single scalar. As Figure 3 illustrated, an aggregate surprise can combine forces that move prices in opposite directions—a rate cut that mechanically raises valuations alongside the unfavorable economic news the action conveys. Collapsing these into one number averages the channels away. Decomposing the surprise into a pure policy shock (MP), a central-bank-information shock (CBI), and a Fed-response-to-news shock

(FRN) lets the *same* characteristic load differently across components.

We include all three policy shocks (MP, CBI, FRN) in one regression rather than running a separate regression for each. Although the components are orthogonal by construction in the full sample, our estimation sample is a truncated subset of announcements—those for which intraday prices and lagged firm characteristics are both available—and this truncation can reintroduce correlation among them. Indeed, in our estimation sample the correlation between the CBI and FRN components reaches 0.50 (Appendix 5.2). Estimating the components jointly purges each interaction $\gamma_{c,k}$ of such contamination from the correlated components; estimating them separately would leave an omitted-component bias.

Finally, we exclude financial firms (SIC 6000–6999) and utilities (SIC 4900–4999), as is standard in the literature, and we lag firm characteristics by one quarter relative to the announcement to avoid look-ahead bias.

| | MP shock | CBI shock | FRN shock |
|---|---|---|---|
| | Coef. | Coef. | Coef. |
| Shock | −0.3449*** | 0.2271*** | −0.2867*** |
| × leverage | −0.0060* | −0.0001 | 0.0038 |
| × tobins q | −0.0110*** | 0.0173** | −0.0195*** |
| × cyclicality | 0.0002 | 0.0080** | −0.0030 |
| × ba spread | 0.0339*** | −0.0416** | 0.0247*** |
| × size | −0.0169*** | 0.0153** | −0.0118 |
| Firm FEs | | ✓ | |
| Clustered SE | | By date | |
| $R^2$ | | 0.570 | |
| $N$ | | 53,442 | |

Table 2: Baseline specification: heterogeneous stock responses to the MP, CBI and FRN shocks, with characteristics demeaned within firm.

Table 2 shows that with all characteristics at their average values, a one standard deviation MP shock reduces stock returns by 0.34 percentage points (p.p.). When a firm is temporarily more leveraged than its own average, the response is 1.73% more negative ($-0.006/-0.34$), reflecting greater exposure to rising borrowing costs. When its Tobin's q is one standard deviation above its own average, the response is 3.2% more negative — a temporarily higher Tobin's Q shifts more of the firm's value to distant cash flows, which a higher discount rate hits harder. Leverage and Tobin's q are the channels of interest here; size and the bid–ask spread enter as controls, confirming faster price adjustment when firms are temporarily larger or more liquid than usual.

For the CBI shock, the main effect is 0.2271: a one standard deviation positive shock raises stock returns by 0.22 p.p. at average characteristics. Tobin's q and cyclicality are significant at 5%. A firm with Tobin's Q one standard deviation above its own average reacts 7.6% more positively — a higher-than-usual Q means more value in distant cash flows, which a stronger-economy signal revises upward. A firm temporarily more cyclical than usual reacts 3.5% more positively (0.0080), as firms tied more closely to economic conditions gain most from favorable Fed signals. The bid–ask spread control is negative ($-0.0416$), consistent with a more subdued within-window response for stocks temporarily less liquid than usual.

For the FRN shock, the main effect is $-0.2867$ — same direction as MP but somewhat smaller, the difference being that FRN reflects the Fed reacting more aggressively to incoming news. Tobin's q is the only significant channel: a firm with Q one standard deviation above its own average reacts 6.8% more negatively ($-0.0195$). The same tilt toward distant cash flows that lifted the stock under CBI now works against it, as a steeper expected rate path discounts those flows more heavily. The bid–ask spread control (0.0247) implies that a firm temporarily *less* liquid than usual reacts 8.6% less negatively, again reflecting slower price discovery when quotes are wide.

As a robustness check, we re-estimate (1) using the raw characteristics in place of their within-firm deviations (Appendix 5.1). The two channels that identify genuine cross-sectional differences: cyclicality under CBI and Tobin's $Q$ under FRN, remain significant and if anything strengthen, while the within-firm-only effects (leverage under MP, size, and Tobin's $Q$ under MP and CBI) lose significance. The regressors jointly explain 57% of variation in stock returns around the FOMC announcements, a figure reported inclusive of the firm fixed effects.

## 3.2 Effects by industry

To test whether the firm-level channels merely proxy for industry membership, we estimate sector-level heterogeneity in the shock responses. The specification for Panel A of Table 3 is

$$r_{i,t} = \alpha_i + \sum_k \beta_k S_{k,t} + \sum_k \sum_{j \neq \text{base}} \delta_{j,k} \left(S_{k,t} \times D_{i,j}\right) + \sum_k \sum_c \gamma_{c,k} \left(S_{k,t} \times x^c_{i,t}\right) + \varepsilon_{i,t}, \tag{2}$$

where $r_{i,t}$ is the percentage change in firm $i$'s stock price in the 30-minute window around FOMC announcement $t$; $S_{k,t}$ is the policy shock $k \in \{\text{MP}, \text{CBI}, \text{FRN}\}$, normalized to unit standard deviation; $D_{i,j}$ is an indicator equal to one if firm $i$ belongs to GICS sector $j$; and $x^c_{i,t}$ are the two controls (bid–ask spread and size), winsorized and standardized. The terms $\alpha_i$ are firm fixed effects, and standard errors are clustered by FOMC announcement date.

Because sector membership is constant within firm, the sector indicators enter only through their interactions with the shocks: the level effect of belonging to a sector is absorbed by the firm fixed effects. One sector must be omitted as the base category. The main effect $\beta_k$ is then the response of the base sector (*Energy*) to shock $k$, and each $\delta_{j,k}$ is a *contrast*: the difference between sector $j$'s response and the base sector's, so that sector $j$'s total response is $\beta_k + \delta_{j,k}$. Panel B augments (2) with the three firm-characteristic interactions of the baseline specification (leverage, Tobin's Q, cyclicality), testing whether those channels survive once sector-level differences are controlled for.

Panel A shows that sectors respond differently to the same announcement. The sectors selling everyday goods (Consumer Staples, Health Care) and Communication Services react significantly less negatively to a contractionary MP shock than the cyclical sectors: the gap between Consumer Staples and Energy is about 0.13 percentage points per unit-standard-deviation shock, roughly a third of the average response. Differences among the cyclical sectors themselves are not statistically significant. Under the FRN shock the pattern differs: it is the sectors selling postponable goods (Materials, Industrials, Consumer Discretionary, Information Technology) that react significantly more negatively than the base sector, while the everyday-goods sectors are indistinguishable from it.

Panel B shows that the firm-level channels survive sector controls: Tobin's Q remains significant with unchanged signs under all three shocks ($-0.016$, $+0.022$, $-0.032$), cyclicality retains its CBI loading ($+0.009$), and leverage its MP loading ($-0.012$). The heterogeneity we document therefore operates *within* sectors — two firms in the same industry that differ in Tobin's Q or cyclical exposure respond differently to the same shock — rather than through sector composition.

Table 3: Heterogeneous 30-minute stock-return responses to monetary policy shocks, by GICS sector and firm characteristic.

| | MP | CBI | FRN |
|---|---|---|---|
| *Panel A: Sector interactions* | | | |
| Shock (base: Energy) | $-0.3947^{***}$ | $0.2189^{***}$ | $-0.2504^{***}$ |
| × Materials | 0.0046 | 0.0171 | $-0.0443^{*}$ |
| × Industrials | 0.0347 | 0.0145 | $-0.0483^{**}$ |
| × Cons. Discr. | 0.0233 | $0.0382^{*}$ | $-0.0552^{**}$ |
| × Cons. Staples | $0.1280^{***}$ | $-0.0703^{**}$ | 0.0236 |
| × Health Care | $0.1226^{***}$ | −0.0361 | −0.0133 |
| × Info. Tech. | 0.0389 | 0.0186 | $-0.0659^{**}$ |
| × Comm. Services | $0.0734^{***}$ | −0.0097 | −0.0362 |
| × Bid–ask spread (ctrl) | $0.0339^{***}$ | $-0.0251^{***}$ | $0.0251^{***}$ |
| × Size (ctrl) | $-0.0105^{**}$ | 0.0107 | −0.0022 |
| *Panel B: Sector interactions + firm-characteristic channels* | | | |
| Shock (base: Energy) | $-0.4036^{***}$ | $0.2273^{***}$ | $-0.2652^{***}$ |
| × Materials | 0.0088 | 0.0187 | $-0.0416^{*}$ |
| × Industrials | $0.0408^{*}$ | 0.0113 | −0.0372 |
| × Cons. Discr. | 0.0314 | 0.0354 | $-0.0445^{*}$ |
| × Cons. Staples | $0.1439^{***}$ | $-0.0795^{**}$ | 0.0406 |
| × Health Care | $0.1342^{***}$ | −0.0468 | 0.0103 |
| × Info. Tech. | 0.0431 | 0.0111 | −0.0450 |
| × Comm. Services | $0.0791^{***}$ | −0.0146 | −0.0256 |
| × Leverage | $-0.0116^{**}$ | 0.0049 | −0.0029 |
| × Tobin's Q | $-0.0157^{***}$ | $0.0219^{**}$ | $-0.0318^{***}$ |
| × Cyclicality | 0.0011 | $0.0093^{**}$ | −0.0039 |
| × Bid–ask spread (ctrl) | $0.0328^{***}$ | $-0.0235^{***}$ | $0.0231^{***}$ |
| × Size (ctrl) | $-0.0116^{**}$ | $0.0156^{*}$ | −0.0095 |
| Firm FE | | ✓ | |
| Clustered SE | | FOMC date | |
| Sectors | | 8 | |
| $N$ | | 48,565 | |

Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Sample restricted to S&P 500 common stocks (CRSP share codes 10–11); Financials and Utilities excluded. Coefficients are in percentage points per unit-SD shock. Each coefficient is the interaction of a unit-standard-deviation shock with a sector indicator or a (winsorized, standardized) firm characteristic. Sector coefficients are contrasts relative to the omitted base sector (*Energy*), whose response is the shock main effect. Panel A includes sector interactions and the two controls; Panel B adds the three firm-characteristic channels.

# 4 Conclusion

This research paper examined how firm characteristics shape stock price transmission of monetary policy, using high frequency intraday returns around a 30-minute window of FOMC announcements. In order to create our results, we used three different types of historical policy shocks capturing unexpected policy tightening (MP), positive news about the economy (CBI), and the Fed's reaction to incoming news (FRN). The two most robust findings in our research are the interaction term of cyclicality under the CBI shock and Tobin's q under the FRN shock. Both of these terms hold across all three specifications of within firm, levels, and within sector. A firm one standard deviation more cyclical (measured by the sensitivity of its profitability to GDP growth) than average reacts 3.5–5% more positively to a CBI shock, while a firm one standard deviation higher in Tobin's q reacts 7–9% more negatively to an FRN shock, depending on whether the interaction is identified within firm or across firms. Tobin's q also shows the sharpest contrast across components: in the baseline specification, the same firm that gains more under a CBI shock loses more under an FRN shock, with nearly symmetric magnitudes, consistent with its value resting on distant cash flows. The CBI side of this contrast, however, does not survive the levels specification.

These results suggest markets differentiate firms within minutes of the Fed announcement window, based on valuation and cyclical exposure rather than debt. This is noteworthy because leverage is a common central mechanism of monetary policy, however in our sample test leverage was weak compared to the other terms in stock price reaction. One likely reason is that S&P 500 firms are mostly far from financial constraints, and that their debt is predominantly long-term and fixed-rate, so short-term rate surprises have little effect on their actual interest payments.

Our sample is limited to S&P 500 firms, which are large and have strong access to capital markets. The leverage effect could have a stronger role, in smaller and financially constrained firms where debt burdens more directly threaten near term cash flows. The 30-minute window is favorable in isolating the effect of the shocks, but it fails to capture how these price reactions evolve in the longer horizons. It is possible that the muted role of leverage reflects a genuinely limited role for financial constraints in the short run, or simply that leverage-related effects take longer to be priced in than our window allows. Some of the effects tested are sensitive to specification: the Tobin's q interaction effects under the MP and CBI shocks are significant within firm and sector but insignificant in the levels specification. This suggests that the variation of specification changes the economic significance of the interaction effects. We interact each firm characteristic separately with the policy shock; we do not examine potential interactions between characteristics themselves, which could reveal additional heterogeneity. Overall, our results show that in the reaction to the Fed announcement, markets price firms based on how exposed their future cash flows and business cycle sensitivity are to the shock, rather than their balance sheet leverage.

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# 5 Appendix

## 5.1 Robustness: levels specification

Table 4 re-estimates the baseline specification with raw characteristics rather than within-firm deviations. So, in this version, the interaction coefficients are identified from differences across firms rather than within a firm over time. The main effects are the same (MP $-0.3449$, CBI 0.2238, FRN $-0.2871$, all significant at 1%) — centering the characteristics does not shift the average firm's response. Most interaction terms are small. A firm one standard deviation more cyclical than average reacts 5.1% more positively to a CBI shock, and one with Tobin's $Q$ a standard deviation higher reacts 9.3% more negatively to an FRN shock ($-0.0266$) — both larger than in the demeaned specification (3.5% and 6.8%), reflecting substantial cross-sectional differences in firm type. The rest interaction effects are not significant at 5% level. The bid–ask spread control stays significant throughout, reflecting the mechanical liquidity effect. The two channels robust across both specifications — cyclicality under CBI and Tobin's $Q$ under FRN — are therefore the most credible, holding whether identification comes from within-firm or cross-sectional variation.

| | MP shock | CBI shock | FRN shock |
|---|---|---|---|
| | Coef. | Coef. | Coef. |
| Shock | $-0.3449$*** | $0.2238$*** | $-0.2871$*** |
| × leverage | $-0.0067$ | $-0.0001$ | $0.0008$ |
| × tobins q | $-0.0045$ | $0.0142$ | $-0.0266$*** |
| × cyclicality | $-0.0031$ | $0.0115$*** | $-0.0049$ |
| × ba spread | $0.0293$*** | $-0.0216$*** | $0.0211$*** |
| × size | $-0.0047$ | $0.0115$ | $-0.0035$ |
| Firm FEs | | ✓ | |
| Clustered SE | | By date | |
| $R^2$ | | 0.570 | |
| $N$ | | 53,442 | |

Table 4: Levels specification: heterogeneous stock responses to the MP, CBI and FRN shocks, with characteristics in raw levels rather than within-firm deviations.

## 5.2 Shock correlation matrix

| | JK MP | JK CBI | JK FRN | BS Orth. MPS | BK Non-yield |
|---|---|---|---|---|---|
| JK MP | 1.000 | -0.113 | -0.014 | 0.949 | -0.092 |
| JK CBI | -0.113 | 1.000 | 0.497 | 0.228 | 0.020 |
| JK FRN | -0.014 | 0.497 | 1.000 | 0.022 | -0.097 |
| BS Orth. MPS | 0.949 | 0.228 | 0.022 | 1.000 | -0.171 |
| BK Non-yield | -0.092 | 0.020 | -0.097 | -0.171 | 1.000 |

Table 5: Pairwise correlations across monetary policy shock measures.

## 5.3 Bauer–Swanson (2023) orthogonalized surprise

| | Levels | Within-firm Demeaned |
|---|---|---|
| | Coef. | Coef. |
| Shock | −0.2930*** | −0.2928*** |
| × leverage | 0.0046 | 0.0087 |
| × tobins q | 0.0032 | −0.0041 |
| × cyclicality | −0.0051 | −0.0001 |
| × ba spread | 0.0217*** | 0.0168 |
| × size | −0.0024 | −0.0148 |
| Firm FEs | | ✓ |
| Clustered SE | | By date |
| $R^2$ | 0.259 | 0.259 |
| $N$ | 46,805 | 46,805 |

Table 6: Single-shock regression: MPS ORTH, levels vs. within-firm demeaned specification

## 5.4 Boehm–Kroner (2024) non-yield shock

| | Levels | Within-firm Demeaned |
|---|---|---|
| | Coef. | Coef. |
| Shock | 0.1056*** | 0.1079*** |
| × leverage | −0.0011 | −0.0002 |
| × tobins q | 0.0076 | 0.0144 |
| × cyclicality | 0.0062 | 0.0039 |
| × ba spread | −0.0201** | −0.0279* |
| × size | −0.0017 | 0.0034 |
| Firm FEs | | ✓ |
| Clustered SE | | By date |
| $R^2$ | 0.042 | 0.044 |
| $N$ | 48,898 | 48,898 |

Table 7: Single-shock regression: Fed NonYield Shock, levels vs. within-firm demeaned specification